MILITARY SPECIFICATION

CONTROL UNIT, COMMUNICATION SYSTEM C-10414(V)( )/ARC
GENERAL SPECIIICATION FOR

This amendment forms a part of Military Specification MIL-C-49227A(AV), dated 31 July 1986, and is approved for use within the US Army Aviation Systems Command, US Army Avionics Research and Development Activity, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1 Add "STANDARDS" above "FEDERAL"

PAGE 29

4.10.8 Delete line 4 "Set the output level..." and subsitute "Observe that the output across pins J1-N and J1-P is 0.39 volts ± 10%."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletion) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

The attached insertable replacement pages listed below are replacements for the stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement pages	Pages replaced
2, 29	2, 29
1, 30	Reprinted without change

Custodian:
Army (AV)

Preparing Activity
Army (AV)
(Project 5821-A188)

AMSC N/A

FSC 5821

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MILITARY SPECIFICATION

CONTROL UNIT, COMMUNICATION SYSTEM C-10414(V)( )/ARC

This specification is approved for use by the Army Aviation Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of control which provides control of inter-communication, audio levels, switching of airborne radio sets and navigation devices and is designated Control Unit, Communication System C-10414(V)( )/ARC. (See 6.1 and 6.4.)

2. APPLICABLE DOCUMENTS. The following document, of the issue as outlined in paragraph 2.1 shall form a part of this specification to the extent specified herein.

2.1 Issues of documents. The effective issue or revision of the following documents shall be that which is in effect on the date of invitation for bids or request for proposal, unless a specific issue is set forth in the cited specifications or contractual documentation.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
PPP-T-0097	Tape, Packaging/Industrial, Filament Reinforced.
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Avionics Research & Development Command, ATTN: SAVAA-C, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD-Form 1426) appearing at the end of this document or by letter.

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FEDERAL (Cont)

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall.
PPP-S-760	Strapping, Nonmetallic (and Connectors).
QQ-S-781	Strapping, Steel, and Seals.

MILITARY

MIL-P-116	Preservation Methods of.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-F-7179	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapons Systems.
MIL-P-7788	Panels, Information, Integrally Illuminated.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-S-19500	Semiconductor Devices, General Specification for.
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for.
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.
MIL-L-25467	Lighting, Integral, Red, Aircraft Instrument, General Specification for.
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packing.
MIL-P-55110	Printed Wiring Boards, General Specification for.

★ STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment.
FED-STD-595	Colors.

MILITARY

MIL-STD-105	Sampling Procedures and Tables of Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.

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4.10.5 Distortion microphone amp.

4.10.5.1 Distortion microphone amp ni voltage. Apply 4.0 volts at 350 Hz and 4.5 kHz to pins J1-L and J1-M through a 5000-ohm to 5-ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control to maximum audio output position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion to determine compliance with 3.13.5.

4.10.5.2 Distortion microphone amp, additional. Apply 0.6 and then 4.0 volts at 1.0 kHz to pins J1-L and J1-M through a 5000 ohm to 5 ohm pad and set selector switch to ICS position, place all monitor switches to OFF, master volume control maximum CW and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and measure the microphone amplifier output distortion at each of the input voltages at 350 Hz, 1000 Hz and 4500 Hz to determine compliance with 3.13.5. Reduce the input voltages to .085 volts at 1.0 kHz and measure the zero crossing distortion to determine compliance with 3.13.5.

4.10.6 Attack and release time, AGC. Set selector switch to ICS position and MIC switch to position 1 (5 ohms) and HOT MIC switch to HOT MIC position and apply 1.34 volts at 1.0 kHz to a 5000 ohm to 5 ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S for a reference level. Remove the input signal for 30 seconds and reapply the input signal voltage, measure the AGC attack time required for the output voltage to reach within 1.0 dB of the reference level. Lower the input signal voltage to .85 ± 10% v then reduce voltage by 16 ± 1 db and hold for 20 seconds. Measure output voltage for a reference level. Increase the input voltage by removing the 16 db attenuation. Hold for 5 seconds and then insert the attenuation. Measure the release time required for the input voltage to reach within 1 db of the reference voltage. These measurements shall be made to determine compliance with 3.13.6.

4.10.7 Breakpoint and slope, AGC. Set MIC switch to position 1 (5 ohms) and selector switch to ICS position and apply 0.1 volt at 1.0 kHz to a 5000-ohm to 5 ohm pad connected to pins J1-L and J1-M and measure the output voltage across pins J2-R and J2-S. Increase the input voltage to 0.2 volt thence to 3.0 volts and measure the output voltages. Compare the input-output voltages to determine the breakpoint. Utilizing the output voltage obtained with 0.5 volts input as a reference level, increase the input signal level to 5.0 volts and measure the output voltage, compare the input-output voltages to determine the slope. Switch the microphone selector switch to the 150 ohm position and repeat above with 2.4K to 150 onm pad with blocking capacitor, 47 ufd or greater connected to pins J1-L and J1-M. The comparisons shall be made to determine compliance with 3.13.7.

* 4.10.8 Transmit audio output, microphone amp. Apply 0.6 volts at 1.0 kHz to a 5,000 ohm to 5 ohm pad connected to pins J1-L and J1-M and set selector switch to position 1, also set MIC switch to position 1 (5 ohms). Connect Pin J1-d to ground. Observe that the output across pins J1-N and J1-P is 0.39 volts ±10%. Measure the output voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-q, with selector switch set to positions 2, 3, 4, and 5. Set microphone switch to position 2, and connect a 2.4K to 150 ohm pad with blocking capacitor, 47 ufd or greater to pins J1-L and J1-M. Apply 0.6 volts at 1.0 kHz to the pad, set selector switch to position 1, and jumper pin J1-d to ground. Observe that the output across pins J1-N and J1-P is 0.39 volts ±10%. Measure the output

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voltage at pin pairs J1-S and J1-T, J1-V and J1-W, J1-k and J1-s, and J1-p and J1-q, with selector switch set to positions 2, 3, 4, and 5 to determine compliance with 3.13.8.

4.10.9 Audio, headset amp.

4.10.9.1 Audio output, single tone, headset amp. Set MIC switch to position 1 (5 ohms), set channel select switch to ICS. Adjust all volume controls to maximum clockwise and push all monitor switches in (On). Connect a 1.0 kHz input at 2.75 volts to pin pair J2-A,B. The audio output shall be 1.25 volts ± 5% at pin pair J2-D,E across 8 ohms. Apply a 3.9 volt and then a 5.4 volt input signal at 1.0 kHz to determine compliance with limiting threshold and maximum limited output requirements of 3.13.9. Reset input level until output is 1.25 volts RMS and then set MIC switch to position 2 (150 ohms). The audio output shall be 1.25 volts ± 10%. Return MIC switch to position 1 (5 ohms) and set input level to 2.75 volts. Then apply input in turn to following pin pairs:

J2-V,W	J2-X,Y	J2-F,G	J2-n,p	J2-J,K	J2-d,e	J2-q,r
J2-U,j	J2-k,m	J2-a,b	J2-c,H	J2-s,t	J2-L,M	J2-N,f

The audio output shall be 1.25 volts ± 5% for each of above inputs. Apply 0.39 volts RMS to pin pair J2-R,S to obtain same output and then set channel select switch to PVT and apply 0.39 volts RMS to pin pair J2-g,p to obtain same 1.25 volts ± 5% output.

4.10.9.2 Audio output, 2-tone test, headset amp. Apply a 5.4 volt input signal of 1.0 kHz to a selected input and a 2.75 volt 1.5 kHz signal to a selected nonrelated input. Observe that the 1.5 kHz output signal is not reduced in amplitude by greater than 9 dB by the presence of the 1.0 kHz signal to determine compliance with 3.13.9.

4.10.10 Response, headset amp. Set MIC switch to position 1 (5 ohms) and apply a 1.0 kHz input at 0.39 volts to pins J2-R and J2-S and adjust volume control for a 0.7 volt output across pins J2-D and J2-E and using this output level as a reference level while maintaining a constant input level, measure the frequency response from 350 to 4,500 Hz and 87 Hz and 18,000 Hz to determine compliance with 3.13.10.

4.10.11 Distortion, headset amp.

4.10.11.1 Distortion, 350 Hz, headset amp. With MIC switch in position 1, volume control fully CW and all monitor controls in (ON) and fully clockwise apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio audio input in turn and measure the headset amp output distortion at 350 Hz for compliance with 3.13.11.

4.10.11.2 Distortion, full range, headset amp. Apply 0.39 volts to ICS and PVT audio inputs and 2.75 volts to each radio channel audio input in turn to obtain 1.25 volts ± 5% output at 1.0 kHz across pins J2-D and J2-E. Measure the headset amp output distortion at 350 Hz, 1000 Hz and 4500 Hz for compliance with 3.13.11. Reduce the input voltage to .055 volts for ICS and PVT audio inputs and to 0.39 volts for each radio channel audio input and measure the zero crossing distortion to determine compliance with 3.13.11.

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